

Sampling methods

- Most **multiparameter** Bayesian models do not permit easy *analytical* calculation of the posterior distribution(s)
- This is not a problem in practice because
 - A suite of computational (numerical sampling) methods are available
 - A suite of software packages for implementing such methods:
 - **JAGS**, OpenBUGS, Stan, NIMBLE, brms (R package),...

General approach of MCMC simulation

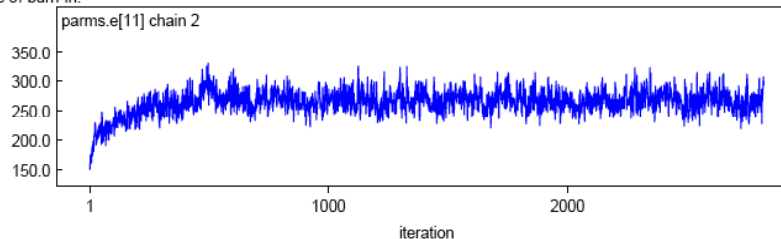
MCMC is a step-by-step mathematical method used by computers to estimate complex probabilities. It works by generating a long sequence of data points (samples) to find an answer ("target" distribution).

- **Step-by-Step Sampling:** The system draws new samples one at a time in a sequence.
- **The "Markov Chain" Rule:** When picking the next sample (iteration j) the choice depends *only* on the most recent sample (iteration $j-1$).
- **Finding the Target:** As the process repeats over many steps, the method's guesses improve. Eventually, the sequence of samples settles down (converges) to accurately represent the final, complex probability (the "target" distribution) you are trying to measure.

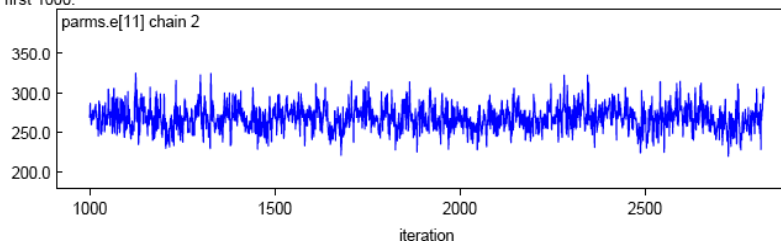
Burn-in

- **Definition:** The initial sequence of samples generated at the very beginning of the MCMC process.
- **The Issue:** Because the chain starts at a more-or-less random initial value, the earliest samples are heavily biased by that starting point and do not accurately represent the final target distribution.
- **The Solution:** You systematically discard this initial batch of iterations (the "burn-in period") so they do not skew your final calculations.

Example of burn-in:

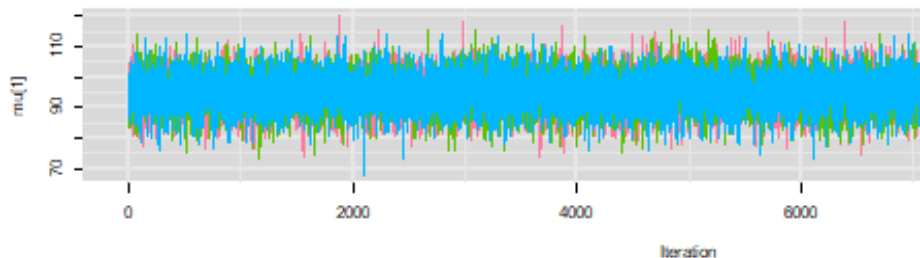


Discard first 1000:



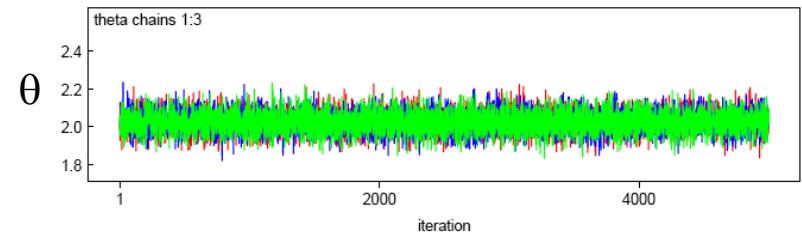
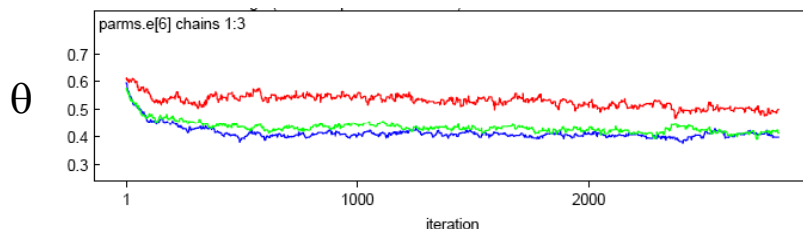
Mixing

- **Definition:** How efficiently and thoroughly the chain explores the entire range of possible values within the target distribution.
- **Good Mixing:** The chain moves freely and rapidly across the entire high-probability space. It explores all relevant areas without getting stuck, meaning you can trust the results with fewer total samples.
- **Poor Mixing:** The chain takes very small steps or gets temporarily trapped in specific areas. This requires you to run the process for significantly more iterations to ensure the entire distribution is accurately mapped.



Convergence

- **Definition:** The point at which the Markov chain has stabilized and is reliably drawing samples from the true target distribution.
- **How to tell:** Once convergence is reached, the overall statistical properties of the sequence (such as the mean and variance) stop shifting and remain stable over time. Can assess visually or with Rhat values (measurement of within- and between chain variance; want values to be < 1.1)
- **Why it matters:** An MCMC model must reach convergence before the data is useful. If the chain has not converged, your samples do not represent the correct probabilities.



Deviance

- **Definition:** A mathematical score that acts as an index of "model fit," telling you how well your statistical model aligns with your actual data.
- **Monitoring in JAGS:** Within the JAGS software, deviance is already a "built-in" logical node. This means you do not have to write a custom equation for it; you can simply tell the program to "monitor" or track it while the model runs.
- **Setup in R:** If you are running JAGS through an R workflow, you must load the Deviance Information Criterion (DIC) tool *before* you initialize your JAGS model. You do this by running the specific command: `load.module("dic")`.